

Executive Summary: Angular Manifold Routing

Modern AI models require enormous computational resources. Much of this cost comes from performing large amounts of computation that may not always be necessary. Angular Manifold Routing is a new approach designed to make AI computation significantly more efficient by routing similar inputs to shared compute paths.

Key Idea

Token embeddings are normalized so they lie on a unit hypersphere and are routed purely by direction rather than magnitude. When the hypersphere is divided into angular sectors, inputs naturally cluster into far fewer sectors than expected.

Observed Scaling Law

$$E(K) \approx c \cdot K^{0.572}$$

Where K is the number of routing sectors available and E(K) is the number actually used. Because the exponent is less than 1, compute grows sublinearly with routing capacity.

Implications

Benefit	Observed Result
Effective compute reduction	3–5× improvement
Cache efficiency	≈2.5× fewer cache misses
Routing scaling	Sublinear growth

If validated at production scale, this architecture could allow large AI systems to run faster and more efficiently while using significantly less hardware.